PROJECT - 5

(Data Analyst Internship)

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Project Title:	Telecom Sector Analysis
Duration:	16th July 2025 – 19th July 2025
Focus:	Exploratory Data Analysis and Power BI Dashboard Development
Domain:	Telecom Industry Analysis
Tool Used:	Power BI , Excel , MySQL
Language:	SQL

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INTRODUCTION

The telecom industry operates in a highly competitive environment where customer churn and revenue optimization remain key challenges. In such a scenario, making informed, data-driven decisions has become essential. This project leverages structured telecom customer data to uncover patterns related to customer retention, payment behavior, service adoption, and revenue distribution using **SQL** and **Power BI**.

The analysis aims to derive actionable insights that can assist telecom businesses in boosting customer satisfaction, increasing profitability, and reducing churn. With SQL handling the backend analytics and Power BI offering an interactive dashboard for data storytelling, the project provides a comprehensive framework for understanding customer behavior and driving smarter decisions.

OBJECTIVE

The primary objective of this project is to perform in-depth data analysis on telecom customer records to:

- Analyze churn patterns and their relation to tenure, contract type, and payment method
- Discover high revenue-contributing services
- Identify segments at higher risk of churn
- Design and deliver an interactive Power BI dashboard that communicates KPIs and business insights clearly

The integration of MySQL and Power BI demonstrates the power of combining backend analytics with interactive visualizations for business reporting and strategic planning.

DATASET OVERVIEW

Filename: cleaned_telecom_data.csv

Records: 7,043 rows

Fields: 21

Key Columns:

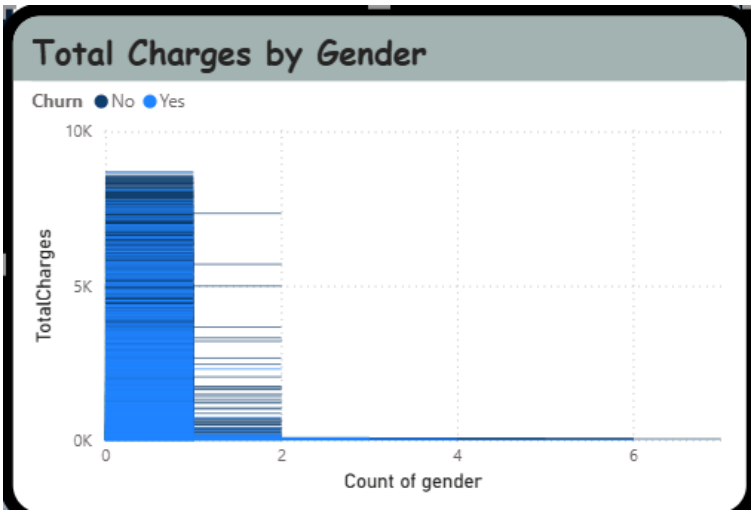
Column Name	**Description**
Customer_id	Unique identifier for each customer
gender	Gender of the customer
Senior citizen	1 = Yes, 0 = No
Partner	Whether the customer has a partner (Yes/No)
Dependents	Whether the customer has dependents (Yes/No)
Tenure	Months the customer has stayed with the company
Phone service	If the customer has phone service
Multiple lines	If customer has multiple phone lines
Internet service	DSL / Fiber optic / No
Contract	Month-to-month / One year / Two year
Paperless billingg	Yes / No
Payment method	Mode of payment
Monthly charges	Monthly bill amount
Total charges	Total charges to date
Churn	Target variable – whether the customer left or stayed

SQL – BASED DATA INSIGHTS

All SQL queries were executed in **MySQL Workbench**, and result sets were either exported or connected live to Power BI for visualization.

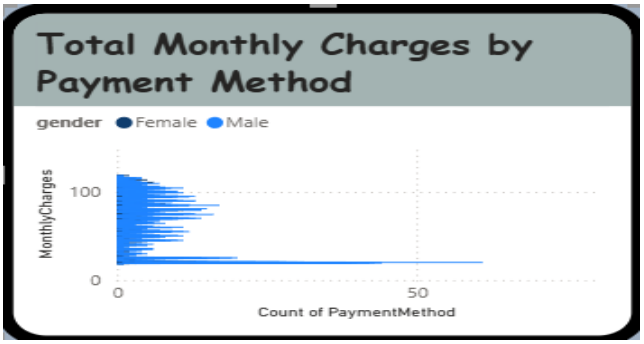
1. Total Charges by Gender

```
SELECT gender, SUM(totalcharges) AS TotalCharges
FROM telecom_data
GROUP BY gender;
```



2. Total Charges by Payment Method

```
SELECT paymentmethod, SUM(totalcharges) AS TotalCharges
FROM telecom_data
GROUP BY paymentmethod;
```

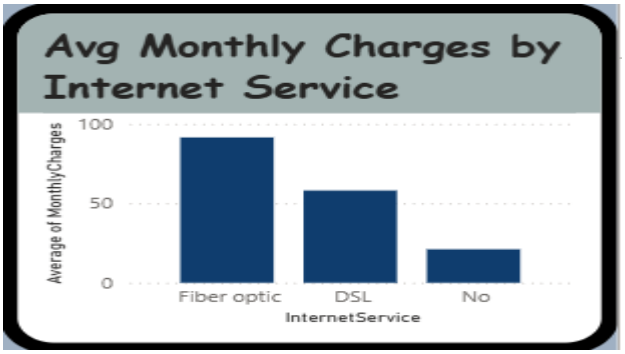


3. Average Monthly Charges by Internet Service

SELECT internetservice, AVG(monthlycharges) AS AvgMonthlyCharge

FROM telecom_data

GROUP BY internetservice;

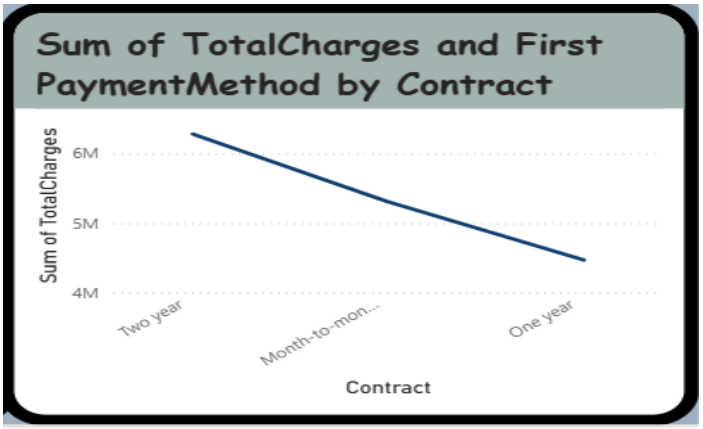


4. Total Charges by Contract Type

SELECT contract, SUM(totalcharges) AS TotalCharges

FROM telecom_data

GROUP BY contract;

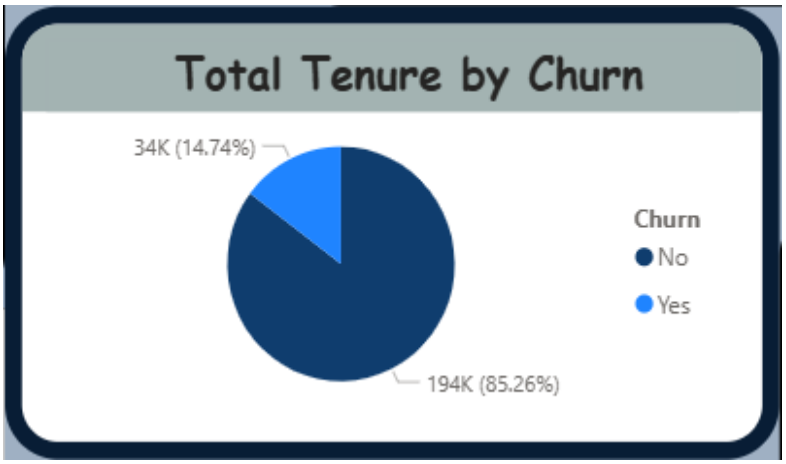


5. Tenure by Churn

SELECT churn, SUM(tenure) AS TotalTenure

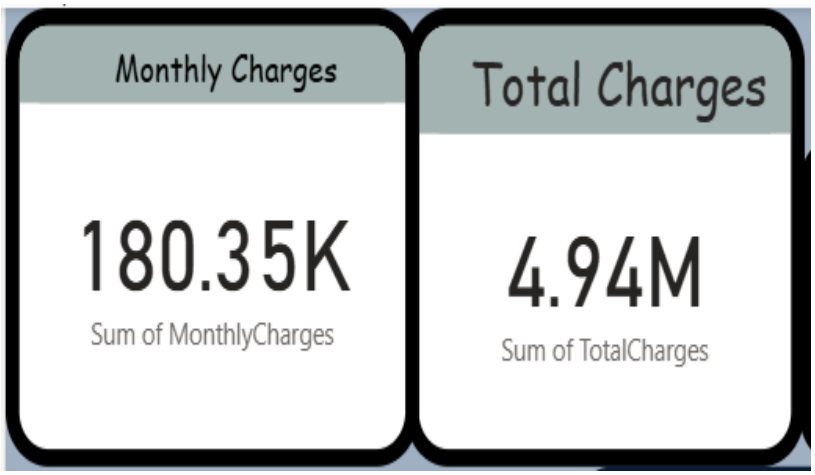
FROM telecom_data

GROUP BY churn;



6. Total KPI Values

```
SELECT
    SUM(monthlycharges) AS TotalMonthlyCharges,
    SUM(totalcharges) AS TotalCharges
FROM telecom_data;
```



These queries helped generate key KPI indicators and customer segments, directly powering the dashboard.

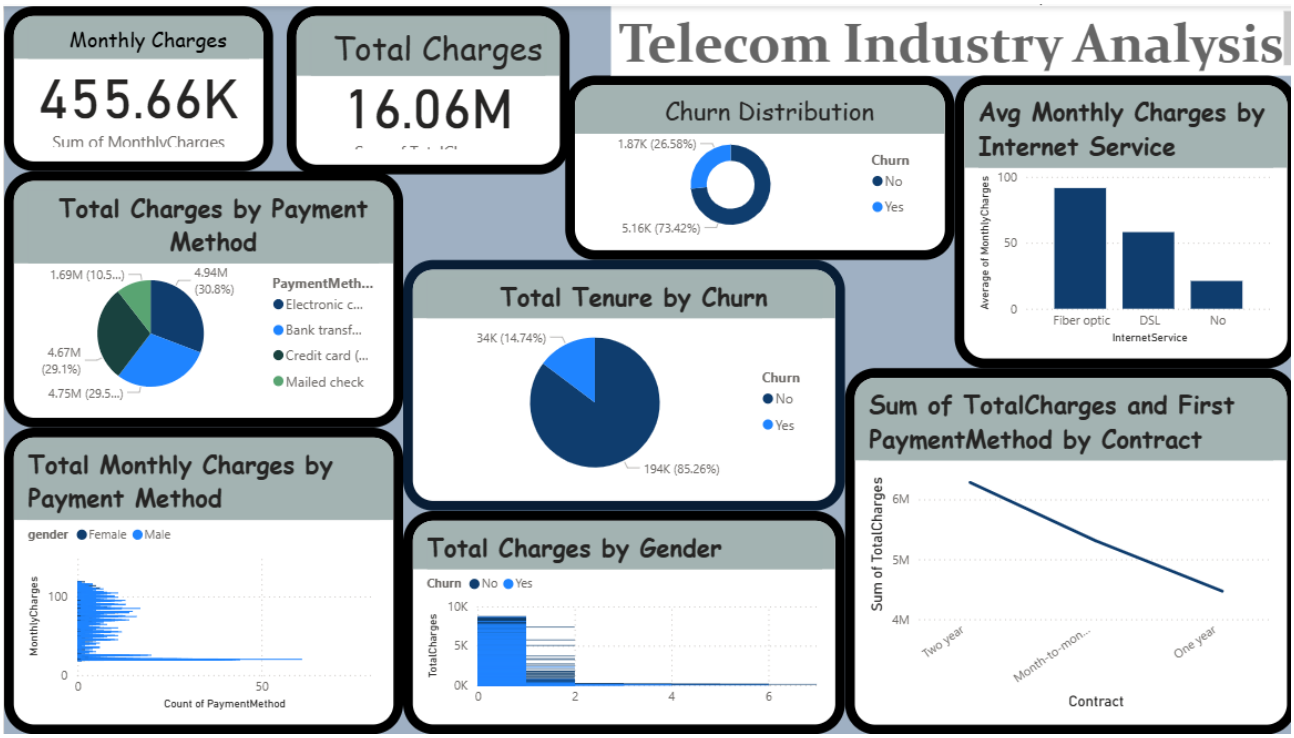
POWER BI DASHBOARD

Tool Used: Power BI Desktop (.pbix)

The Power BI dashboard served as the central visualization layer, converting SQL-based results into business-friendly visuals. Key components include:

- **Scorecards:** Total Charges and Monthly Charges
- **Pie Charts:** Churn distribution, Tenure by Churn
- **Bar/Column Charts:**
  - Total Charges by Gender
  - Monthly Charges by Payment Method
  - Total Charges by Contract Type
  - Average Monthly Charges by Internet Type

Each visual was formatted with borders, titles, color palettes, and background themes for better storytelling and aesthetics.



KEY BUSINESS FINDINGS

1. **High Churn Among Month-to-Month Users**
   Users on month-to-month contracts show the highest churn rate.
   - *Recommendation:* Offer loyalty rewards for converting to long-term plans.

2. **Electronic Check Users Are More Likely to Churn**
   They form a large part of the churn segment.
   - *Recommendation:* Encourage switch to stable payment modes like auto-debit.

3. **Fiber Optic Customers Pay the Most But Churn More**
   High-paying users also exhibit dissatisfaction.
   - *Recommendation:* Prioritize service quality and tech support.

4. **Long Tenure Indicates Loyalty**
   Churn is low among long-tenured users.
   - *Recommendation:* Early engagement and support improve retention.

5. **Senior Citizens Have Higher Churn Rates**
   Possibly due to less tech familiarity.
   - *Recommendation:* Offer customized, simplified service bundles.

6. **Streaming Services Correlate with Retention**
   Users with TV/movie services churn less.
   - *Recommendation:* Promote streaming bundles for engagement.

7. **80/20 Revenue Pattern Confirmed**
   A few segments contribute the majority of revenue.

Recommendation: Focus retention efforts on top-paying groups.

8. Auto-Payment Users Bring Stable Revenue

Bank transfers and credit cards ensure steady income.

- *Recommendation:* Incentivize customers to switch to auto-pay.

CONCLUSION

This internship project offered a hands-on opportunity to apply SQL and Power BI to a real-world telecom dataset. By analyzing service preferences, payment behavior, contract types, and churn metrics, we derived insights that are valuable for both operational efficiency and strategic growth.

Key findings—like churn risks in electronic check users, the impact of tenure, and opportunities to enhance loyalty through bundling—highlight how data analytics can guide revenue-focused decisions.

This project not only improved my technical proficiency in SQL and dashboarding but also strengthened my ability to convert data into real business impact.

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End of Report